



Course Content

Introduction to hydrosphere, atmosphere, geosphere and anthrosphere (tangible human interaction space), with chemical interactions, deterioration (pollution and degradation/depletion) and remediation processes etc., understanding of our environment its chemistry, current industrialized ecology, environmental biotechnology, emphasizing biotechnological tools, such as microbial, genetic, and biochemical mechanisms, pollution, and remediation, sustainable biotechnological interventions in waste treatment, bioenergy production, and pollution control, integration of biotechnology with environmental systems.

In addition to part I (General handout for all courses appended to the timetable) this portion gives further details regarding the course.

Course No. : ENVS F212

Course Title : Environmental Chemistry and Environmental Biotechnology

Instructor-in-charge and other instructor : Prof. Bibhas R. Sarkar (IC), and Prof. B. Vani

Course Description:

The course will provide an overview of the different aspects of the environment and the chemistry integrally involved in it. Broad description of the constituents of the environment namely the hydrosphere, the atmosphere, the geosphere and the anthrosphere (tangible human interaction space) will be provided with emphasis on the chemical interactions involved in each of them in their respective formation, balanced existence, deterioration (pollution and degradation/ depletion) and remediation processes etc. The course will be essential to gaining an understanding of our environment and the chemistry behind the processes occurring within, with relevance to the current industrialized ecology. The course will also provide an overview of environmental biotechnology, emphasising biotechnological tools, such as microbial, genetic, and biochemical mechanisms involved in ecosystem function, pollution, and remediation. The course will cover sustainable biotechnological interventions in waste treatment, bioenergy production, and pollution control. It is designed to build a foundational understanding of how biotechnology integrates with environmental systems, enabling learners to address environmental degradation in the context of modern industrialized ecology.

Learning Objectives

The course will be broadly divided based on the three major components as hydrosphere, atmosphere and geosphere. Learning objectives for each of the broad topics as discussed in the course plan has been described along with – what the learner should be able to do by studying these.

Text Books

T1. **Principles of Environmental Chemistry**, Second Edition, James E. Girard, Jones & Bartlett Learning, Second Edition 2010





T2: Alan Scragg, *Environmental Biotechnology*, Oxford University Press, 2009 reprint.

Reference books:

R1. *Environmental Chemistry* by Stanley Manahan, 9th Edition, CRC Press, 2010

R2. A K Chatterjee, *Introduction to Environmental Biotechnology*, PHI, India, 2000.

R3: Michael T. Madigan, Kelly S. Bender, Daniel H. Buckley, W. Matthew Sattley and David A. Stahl, *Brock Biology of Microorganisms*, 14th Edition, 2017, Pearson International Edition

R4: Raina M Maier, Ian L Pepper and Charles P Gerba, *Environmental Microbiology*, 2nd Ed., Academic Press, 2009.

R5: Howard S Peavy, Donald R Rowe, George Tchobanoglous, *Environmental Engineering*, McGraw-Hill International Editions, Civil Engineering Series, 1985.

R6: Bimal C. Bhattacharyya and Rintu Banerjee, *Environmental Biotechnology*, Oxford Higher Education, 2007.

R7: Godfrey Boyle, *Renewable Energy* – Power for a sustainable future, 2nd Ed, Oxford, Indian Edition, 2011 reprint.

R8: Rehm H J and Reed G, *Biotechnology, a comprehensive treatise*, VCH Verlag, Germany, 1999.

Course Plan:

Topic	Learning Objectives	Lecture Nos.	Ref. text
Planet Earth: Rocks, Life and Energy	Formation of Universe, Different Layers, Formation of Oceans, Atmosphere, Ores and Metals, Origin of Life, Introduction to Atmosphere and its components	1-3	Ch1-3
Chemistry of Troposphere and Stratosphere	Chemical Reactions in Atmosphere, CO, NOx, VOCs, Automobile Pollutants, SOx, Smog, Temperature Inversion, Dobson Unit, Ozone Production and Destruction, Depletion of Ozone Layer, Air pollution & Analysis (ATMOS, LIDAR etc.)	4-6	Ch5-7
Chemistry in Hydrosphere	Distribution of Water on Earth, Composition, Hydrologic Cycle, Unique Properties of Water, Acid Rain, Water Use and Shortage, Water Management and Conservation, Water Withdrawal, Consumption and Pollution – types of pollutants, Chemistry of water analysis and treatments	7-9	Ch8-10
Fossil Fuels and Alternative energy	Chemistry in energy generation – Fossil fuels, nuclear energy and alternative energy	10-11	Ch11, 13





Inorganic Chemicals in Environment	Bioaccumulation of Heavy Metals (Lead, Mercury, Cadmium, Arsenic), Asbestos materials, classification, characteristics	12-14	Ch14,19
Organic Chemicals in Environment	Polychlorinated Hydrocarbons, Persistent, Using EPA's PBT Profiler, Analysis of Persistent Organic Pollutants in Environmental Samples. Insecticides - Organophosphates, Carbamates, Systemic Herbicides, Analysis, Alternative Methods. Types, Exposure, Dose and Response, Teratogens, Mutagens and Carcinogens, Nucleic Acids, Genetic Code	15-17	Ch15, 17,18
Green Chemistry and Sustainability	Sustainable Consumption and Production, Five Axioms, UN Commission on Sustainable Development, Life Cycle Thinking, Leapfrog Technology, Green Chemistry.	18-20	Ch16
Foundations of Environmental Science and Biodiversity	Environmental issues and sustainable development frameworks, ecosystem structure, biodiversity concepts and conservation strategies, genetic variation and molecular tools for biodiversity assessment.	20-22	(1)T
Microbial Physiology, Metabolism, and Genetic Engineering	Microbial cell structure, classification, metabolism, enzyme kinetics, growth models, biofilm dynamics, gene transfer mechanisms and basic biochemistry, genetic engineering tools, biotechnology ethics, biosafety.	23-25	R3
Wastewater and Industrial Effluent Treatment	Primary, secondary, and tertiary wastewater treatment processes, hybrid, and advanced reactor systems. Treatment of industrial effluents using microbial and biotechnology approaches.	26-28	4(T); 4(R2); 20 (R4); 5(R5)
Bioenergy and Bioconversion Technologies	Anaerobic digestion, UASB, and sludge management for biogas generation, microbial pathways for biohydrogen, biomethane, and bioethanol production. Downstream processing and value-added bioproduct recovery. Biofertilizers, vermiculture, organic farming, and nitrogen fixation.	29-31	7(T); 10(R2); 4(R7); 3(R6)
Bioreactors and Environmental Process Design	Types of bioreactors and microbial fuel cells for bioenergy and wastewater treatment, syngas production and gasification systems. Design principles for WTP/STP components, clarifiers, and elevated reservoirs under load conditions.	32-34	R8;6(R5)
Agricultural Biotechnology and Eco-treatment Systems	Use of bioinsecticides, biofungicides, and PGPRs to improve agricultural productivity, symbiotic systems like mycorrhiza and Rhizobium, natural eco-treatment systems using wetlands, hyacinth, and vegetative filters for water/land remediation.	35-37	9(T)





Bioremediation, Pollution Control, and Environmental Monitoring	Microbial, Phyto, Bio based remediation strategies of metals, xenobiotics, and odorous compounds. Biosensors, metagenomics, and bioprospecting tools for monitoring. Climate change mitigation, carbon sequestration, and bioenergy commercialization with regulatory considerations	38-40	5(T); 7(R2); 20(R4); R1 6(R2); 21(R4)
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Evaluation Scheme:

Sl. No.	Evaluation Component.	Duration	Weightage, %	Date, Time & Venue.	Nature of Component.
1	MIDSEM Test	1.5h	25	07 Oct 2026, FN1	CB + OB
2	Tutorial Tests	-	30	To be declared	
3	Comprehensive Examination	3h	45	08 Dec 2026, FN	CB + OB

Chamber Consultation Hour: To be announced in class

Notices: All notices will be put up in Nalanda

Minimum criterion for awarding valid grades: Where a registered student did not give the instructor inadequate opportunity to evaluate him/her by absenting himself/herself from quizzes/ tests/ examinations/other components of evaluation, or by appearing in the same for the sake of appearance without applying himself/herself to the task in hand or by submitting a blank answerscript, this event will be reported as NC (Not Cleared).

A student should obtain 30% of the average of the top 10 performers in the class, or 40% of the median marks of the class, whichever is lower to clear the course. If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC.

Make-up Policy: As per institute's policy

Instructor-in-charge

